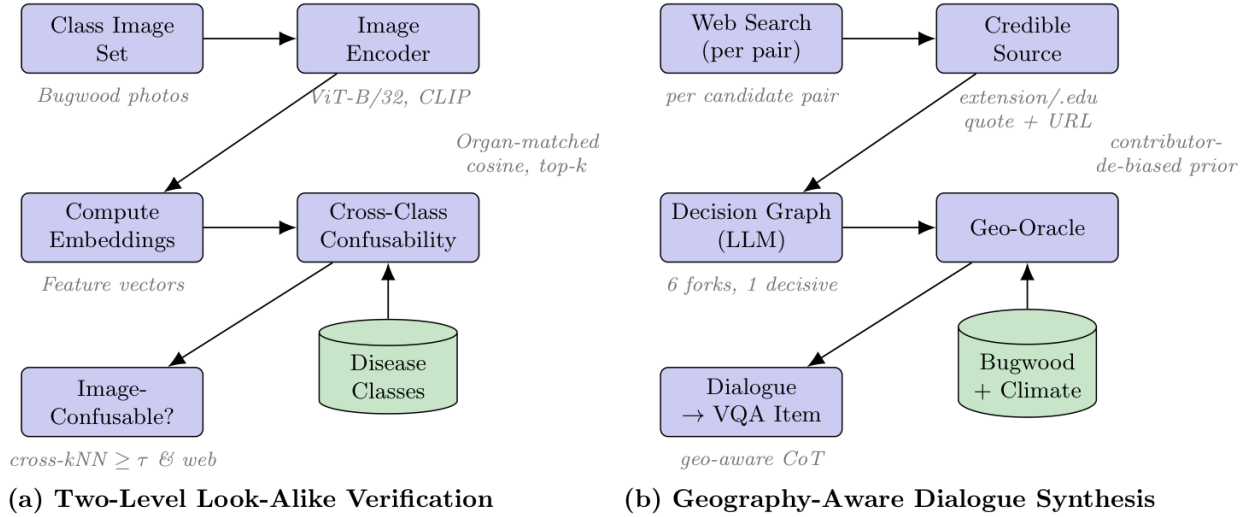


GeoPhyto-QA

Geography-aware look-alike diagnosis VQA from Bugwood imagery

Two-level (CLIP image $\wedge$ web-verified) look-alike confirmation $\cdot$ decision-graph CoT $\cdot$ farmer-expert dialogue

What this is. Each item is a multi-turn, image-grounded farmer $\leftrightarrow$ agricultural-expert dialogue where the expert reasons through a *look-alike decision graph* to tell apart two confusable diseases on the same crop, and the reasoning is *geography-aware* (region/season/local disease pressure from a contributor-de-biased oracle drive the answer). A pair becomes a look-alike only if it is BOTH visually entangled (CLIP, organ-matched) AND documented as confused by a credible web source. **Latest build. 465** dialogue items over **19** confirmed look-alike pairs. Two reasoning lanes: **118** *image-decisive* (Lane A — the deciding sign is visible, so the answer is read from the photo and geography is a swap *control*) and **347** *image-ambiguous* (Lane B — the sign is absent, so a contributor-de-biased regional prior breaks the tie and the location swap is *expected to flip* the answer). 118 counterfactual controls. Splits: test_heldout_region 28, test_heldout_species 203, test_random 26, train 208.



Pipeline: (a) two-level look-alike verification; (b) geography-aware dialogue synthesis.

Confirmed look-alike pairs (image $\wedge$ web): 23

Each pair passed CLIP organ-matched cross-kNN entanglement *and* carries a verbatim quote from a credible source. (Sweep in progress; count grows.)

Crop	Look-alike pair	kNN	Source
Corn	Common Corn Rust <i>vs</i> Southern Corn Rust	0.31	Southern Rust and Common Rust on Hybrid Co
Corn	Corn Smut <i>vs</i> Fusarium Graminearum Schwabe	0.24	Scout for Corn Molds and Ear Rots This Har
Corn	Anthrachnose <i>vs</i> Gray Leaf Spot	0.20	Anthrachnose in Corn Crop-Watch Universi
Cucumber	Anthrachnose <i>vs</i> Corynespora Leaf Spot	0.29	Corynespora cassicola
Cucumber	Angular Leaf Spot Of Cucumber <i>vs</i> Cucurbit Downy Mildew	0.26	Cucurbit Downy Mildew NC State Extension
Cucumber	Anthrachnose <i>vs</i> Gummy Stem Blight	0.25	Gummy Stem Blight of Cucurbits NC State

Crop	Look-alike pair	kNN	Source
Cucumber	Corynespora Leaf Spot <i>vs</i> Cucurbit Downy Mildew	0.25	Corynespora cassiicola
Cucumber	Angular Leaf Spot Of Cucumber <i>vs</i> Corynespora Leaf Spot	0.23	Key to Common Problems of Cucumbers Univ
Cucumber	Anthracnose <i>vs</i> Cucurbit Downy Mildew	0.22	Cucurbit Downy Mildew NC State Extension
Cucumber	Angular Leaf Spot Of Cucumber <i>vs</i> Anthracnose	0.21	Anthracnose of Cucurbits NC State Extens
Highbush Blueberry	Anthracnose <i>vs</i> Twig Canker	0.23	FS1361: Anthracnose Fruit Rot in Highbush
Hosta	Hosta Virus X <i>vs</i> Tobacco Rattle Virus	0.21	What's wrong with my plant? Hosta - Spots
Juniper	Kabatina Blight <i>vs</i> Phomopsis Cankers And Twig Blights	0.28	Kabatina and Phomopsis Tip Blight Yard a
Melon	Anthracnose <i>vs</i> Cucurbit Downy Mildew	0.23	Under the Microscope: Melon Growers Should
Soybean	Bacterial Blight <i>vs</i> Bacterial Pustule Of Soybean Disease	0.25	Bacterial Blight and Bacterial Pustule
Soybean	Frogeye Leaf Spot <i>vs</i> Septoria Leaf Spot	0.24	Frogeye Leaf Spot on Soybean Cooperative
Soybean	Bacterial Pustule Of Soybean Disease <i>vs</i> Septoria Leaf Spot	0.23	Soybean Disease Diagnostic Series NDSU A
Soybean	Bacterial Pustule Of Soybean Disease <i>vs</i> Downy Mildew	0.21	Soybean Disease Diagnostic Series NDSU A
Strawberry	Phomopsis Leaf Blight And Fruit Rot <i>vs</i> Ramularia Grevilleana Verkley & U. Braun	0.30	PP359: Leaf Spot Diseases of Strawberry
Tomato	Early Blight <i>vs</i> Verticillium Wilts	0.30	Verticillium Wilt of Tomato — Cornell Coop
Wheat	Tan Spot <i>vs</i> Zymoseptoria Tritici	0.25	Fungal, Bacterial, and Physiological Leaf
Wheat	Barley Yellow Dwarf Virus <i>vs</i> Wheat Streak Mosaic Virus	0.23	Barley Yellow Dwarf CropWatch Universi
Wheat	Stripe Rust <i>vs</i> Wheat Leaf Rust	0.23	An Overview of Stripe Rust of Wheat — Crop

Web evidence (verbatim)

- **Juniper: Kabatina Blight vs Phomopsis Cankers And Twig Blights** — “*Tip blight of junipers is caused by two different species of fungi, Phomopsis juniperovora and Kabatina juniperi, which cause similar symptoms in the affected plants.*”
<https://yardandgarden.extension.iastate.edu/encyclopedia/kabatina-and-phomopsis-tip-blight>
- **Wheat: Barley Yellow Dwarf Virus vs Wheat Streak Mosaic Virus** — “*Symptoms of barley yellow dwarf are highly variable and can be confused with those of wheat streak mosaic, nutrient deficiency, root and crown diseases, and environmental stress.*”
<https://cropwatch.unl.edu/plant-disease/wheat/barley-yellow-dwarf/>
- **Cucumber: Anthracnose vs Corynespora Leaf Spot** — “*The spots can be mistaken for insect damage or anthracnose.*”
[https://wiki.bugwood.org/Corynespora_cassiicola_\(target_spot_of_cucurbits\)](https://wiki.bugwood.org/Corynespora_cassiicola_(target_spot_of_cucurbits))
- **Wheat: Stripe Rust vs Wheat Leaf Rust** — “*The pustules occur in a random pattern and can be confused with other rust diseases, such as leaf rust.*”
<https://cropprotectionnetwork.org/publications/an-overview-of-stripe-rust-of-wheat>

Look-alike image pairs (12)

Real Bugwood photos of each member, drawn from a shared organ/presentation bucket so the two are genuinely comparable. Left = member A, right = member B.



A: Common Corn Rust

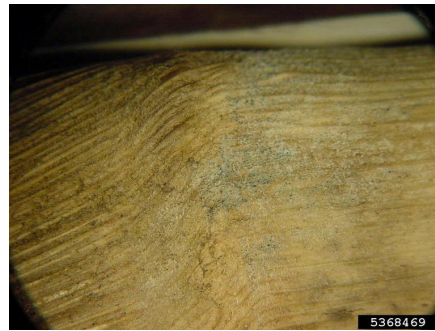


B: Southern Corn Rust

Corn · CLIP kNN = 0.31, bucket=symptom · *“Based on the fact that they both produce rusty looking pustules on the leaves, producers may actually be confusing common rust with southern rust, esp”*



A: Corn Smut



B: Fusarium Graminearum Schwabe

Corn · CLIP kNN = 0.24, bucket=sign · *“Corn smut can be confused with other corn molds and fungal ear rot organisms that can produce dangerous mycotoxins.”*



A: Anthracnose



B: Gray Leaf Spot

Corn · CLIP kNN = 0.20, bucket=symptom · *“Immature lesions are not diagnostic and can easily be confused with Gray Leaf Spot or Eye spot.”*



A: Anthracnose



B: Corynespora Leaf Spot

Cucumber · CLIP kNN = 0.29, bucket=symptom · “The spots can be mistaken for insect damage or anthracnose.”



A: Angular Leaf Spot Of Cucumber



B: Cucurbit Downy Mildew

Cucumber · CLIP kNN = 0.26, bucket=symptom · “Downy mildew is often confused with other foliar diseases such as: *Alternaria* leaf blight, caused by *Alternaria cucumerina*... Angular leaf spot, cause”



A: Anthracnose



B: Gummy Stem Blight

Cucumber · CLIP kNN = 0.25, bucket=symptom · “Gummy stem blight and black rot can look like other plant diseases that affect cucurbits; however, this pathogen can be distinguished by stem and vine”

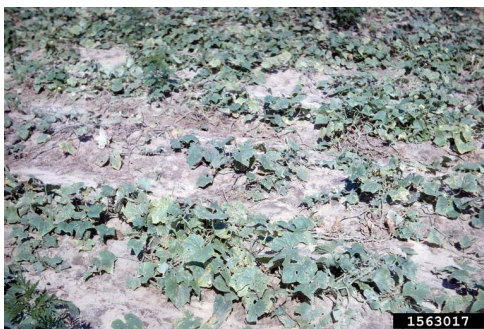


A: Corynespora Leaf Spot



B: Cucurbit Downy Mildew

Cucumber · CLIP kNN = 0.25, bucket=symptom · *“Target spot occurs sporadically and can be confused with downy mildew and other leaf spotting diseases.”*



A: Angular Leaf Spot Of Cucumber



B: Corynespora Leaf Spot

Cucumber · CLIP kNN = 0.23, bucket=symptom · *“Several different fungal and bacterial pathogens cause these symptoms and these diseases may occur simultaneously. It is challenging to tell them apart”*



A: Anthracnose



B: Cucurbit Downy Mildew

Cucumber · CLIP kNN = 0.22, bucket=symptom · *“Downy mildew is often confused with other foliar diseases such as: ... Anthracnose, caused by Colletotrichum orbiculare”*

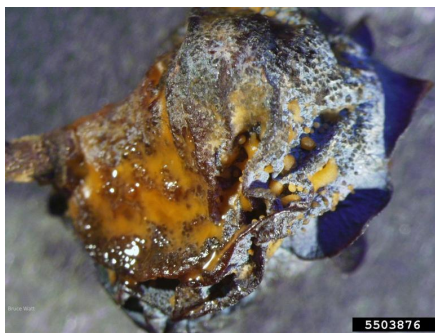


A: Angular Leaf Spot Of Cucumber



B: Anthracnose

Cucumber · CLIP kNN = 0.21, bucket=symptom · “*Anthracnose is often confused with other foliar diseases such as: ... Angular leaf spot, caused by Pseudomonas syringae pv. lachrymans*”



A: Anthracnose



B: Twig Canker

Highbush Blueberry · CLIP kNN = 0.23, bucket=sign · “*These symptoms can often be confused with other fungal diseases of highbush blueberry, including Phomopsis canker and twig blight and Botrytis blossom*”



A: Hosta Virus X



B: Tobacco Rattle Virus

Hosta · CLIP kNN = 0.21, bucket=symptom · “*Hosta Virus X, Tobacco Rattle virus, and others ... Lab test required to distinguish between viruses and to verify infection*”

Sample items (real records, both lanes)

Lane A — image-decisive (sign visible)

gpqa-1573508-a split: test_heldout_species
Anthracnose vs Cucurbit Downy Mildew @ North Carolina (South) · part: leaf

Dialogue

[F1] **Farmer:**  My cucumber leaf — the plant has spots with fuzzy stuff and a yucky ooze on it, which is making it look sick. What is this?

[E1] **Expert:** Looking at your photo, I can see sunken spots that ooze a slimy pink to salmon-orange goo in damp weather. That points to Anthracnose rather than Cucurbit Downy Mildew, which would instead show a fuzzy gray-to-purple downy growth.

[F2] **Farmer:** Will it spread, and what should I do about it?

[E2] **Expert:** Use pathogen-free seed and resistant varieties; rotate out of cucurbits 2-3 years; avoid working in wet fields and overhead irrigation to limit splash; bury or remove crop debris. Apply protectant fungicides (chlorothalonil, mancozeb) on a preventive schedule, rotating in QoI/SDHI chemistries per resistance-management guidelines.

[F3] **Farmer:** My neighbor says it's just the weather and nothing to worry about.

[E3] **Expert:** Weather can play a part, but the sunken spots that ooze a slimy pink to salmon-orange goo in damp weather I can see in your photo is what makes this Anthracnose and not Cucurbit Downy Mildew, so it's worth acting on.

Decision-graph CoT

- **PERCEIVE** (*image (VLM lay observation)*) — On the cucumber leaf: the plant has spots with fuzzy stuff and a yucky ooze on it, which is making it look sick.
- **READ_SIGN** (*image (decisive sign visible)*) — The photo shows sunken spots that ooze a slimy pink to salmon-orange goo in damp weather — the deciding sign.
- **RULE_OUT** (*decision graph*) — Cucurbit Downy Mildew would instead show a fuzzy gray-to-purple downy growth, which is not present.
- **CONCLUDE** (*image*) — From the visible sign this is Anthracnose, not Cucurbit Downy Mildew. Geography is not needed.

Gold: dx = Anthracnose; ruled out Cucurbit Downy Mildew; geography_used = ; answerable_from_image = True.

Decisive cue: Look on the fruit and stems for sunken spots that ooze a slimy pink to salmon-orange goo in damp weather; that pinkish ooze on the cukes is the giveaway, and there's no fuzzy growth underneath the leaves

Counterfactual (control): swap → Iowa. No change — the decisive sign is visible in the photo, so the diagnosis is from the image and location is irrelevant.

Two-level evidence: CLIP organ-matched cross-kNN = 0.2242; web: “Downy mildew is often confused with other foliar diseases such as: ... Anthracnose, caused by Colletotrichum orbiculare” <https://content.ces.ncsu.edu/cucurbit-downy-mildew>

Lane B — image-ambiguous (geography breaks the tie)

gpqa-1563005-a split: test_heldout_species
Anthracnose vs Cucurbit Downy Mildew @ North Carolina (South) · part: leaf

Dialogue

[F1] **Farmer:**  My cucumber leaf — the leaves look sick with spots and some are wilting. I'm farming in North Carolina. What is this?

[E1] **Expert:** From the photo I can see the general problem, but the feature that separates Anthracnose from Cucurbit Downy Mildew — sunken spots that ooze a slimy pink to salmon-orange goo in damp weather vs a fuzzy gray-to-purple downy growth — isn't visible here. Given you're in the South, where anthracnose is well established in the South this season, it's more likely Anthracnose.

[F2] **Farmer:** So what should I do, and how can I be sure?

[E2] **Expert:** To confirm, look on the fruit and stems for sunken spots that ooze a slimy pink to salmon-orange goo in damp weather; that pinkish ooze on the cukes is the giveaway, and there's no fuzzy growth underneath the leaves. In the meantime, manage for Anthracnose: use pathogen-free seed and resistant varieties; rotate out of cucurbits 2-3 years; avoid working in wet fields and overhead irrigation to limit splash; bury or remove crop debris. Apply protectant fungicides (chlorothalonil, mancozeb) on a preventive schedule, rotating in QoI/SDHI chemistries per resistance-management guidelines.

[F3] **Farmer:** A friend thinks it's just fertilizer burn.

[E3] **Expert:** Without the deciding sign in the photo, location is the best guide: in the South anthracnose is well established in the South this season, so Anthracnose is the call until a check of sunken spots that ooze a slimy pink to salmon-orange goo in damp weather says otherwise.

Decision-graph CoT

- **PERCEIVE** (*image (VLM lay observation)*) — On the cucumber leaf: the leaves look sick with spots and some are wilting.
- **SIGN_ABSENT** (*image (decisive sign NOT visible)*) — The deciding sign (sunken spots that ooze a slimy pink to salmon-orange goo in damp weather vs a fuzzy gray-to-purple downy growth) is not visible in this photo.
- **GEO** (*contributor-de-biased TRAIN prior*) — In the South, anthracnose is well established in the South this season (prior weight 12 vs 3 for Cucurbit Downy Mildew).
- **CONCLUDE** (*geographic prior*) — Photo underdetermines it, so geography tips it to Anthracnose; confirm by checking sunken spots that ooze a slimy pink to salmon-orange goo in damp weather.

Gold: dx = Anthracnose; ruled out Cucurbit Downy Mildew; geography_used = True; answerable_from_image = .

Region prior (TRAIN): South favors Anthracnose (12 vs 3).

Decisive cue: Look on the fruit and stems for sunken spots that ooze a slimy pink to salmon-orange goo in damp weather; that pinkish ooze on the cukes is the giveaway, and there's no fuzzy growth underneath the leaves

Counterfactual (expect shift): swap → Colorado. Answer should flip toward Cucurbit Downy Mildew (or to uncertainty): the photo lacks the deciding sign, so a region where Anthracnose is implausible removes the only basis for it.

Two-level evidence: CLIP organ-matched cross-kNN = 0.2242; web: “Downy mildew is often confused with other foliar diseases such as: ... Anthracnose, caused by *Colletotrichum orbiculare*” <https://content.ces.ncsu.edu/cucurbit-downy-mildew>